

From Repeated Corrosion Images to Proxy Remaining Useful Life: An Interpretable Feasibility Pipeline for Ferrocement Specimens

Parastoo

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Corrosion Assessment and Structural Prognosis

Abstract—This paper presents an interpretable, step-by-step pipeline that turns repeated corrosion images of ferrocement specimens into a structured condition-assessment workflow. The study was motivated by a clear imbalance in the available data: visible corrosion labels are available for every image observation, while structural measurements are available only once per specimen at the end of the test program. Rather than claiming direct prediction of true remaining useful life, the study uses a staged formulation that matches the evidence actually present in the dataset. The workflow first reconstructs a specimen-by-time dataset from spreadsheet records and images, then preprocesses each image, extracts interpretable corrosion descriptors, predicts visible corrosion severity, estimates hidden structural condition, fits specimen-level degradation curves, and finally computes the time to future engineering threshold crossing as a proxy remaining useful life. The dataset contains 792 image observations from 48 specimens across two campaigns, while structural measurements are available for only 48 late-stage rows. The image-analysis stage extracts 91 interpretable features describing corrosion amount, location, morphology, color, and texture. In the grouped specimen-holdout benchmark, the best visible-corrosion models achieved mean absolute error of 1.38 for total visible rust and 4.21 for peak rust, improving over the earlier deep-learning baseline on the shared peak-rust task. Hidden-damage estimation remained feasible but much weaker, with mean absolute error of 3.90 for wire-area loss and 0.121 kN for ultimate load. Degradation fitting produced 192 specimen-target curve fits, most of which selected piecewise-linear behavior. The final remaining-life outputs are useful as screening quantities rather than true lifetime predictions. The main contribution of the study is therefore the construction of a coherent and interpretable corrosion-to-condition pipeline, together with evidence that sparse structural supervision is the main bottleneck in moving from visible corrosion to structural prognosis.

Index Terms—corrosion assessment, ferrocement, hidden damage, degradation modeling, proxy remaining useful life, interpretable image features

I. INTRODUCTION

Visible corrosion is often the earliest measurable sign of deterioration. Rust patches, local darkening, and changes in color or texture can be observed directly in images, which makes image-based monitoring attractive for repeated inspection. The engineering goal, however, is not only to describe surface appearance. The larger question is whether visible corrosion

can support conclusions about hidden structural condition, future deterioration, and remaining service margin.

That question becomes difficult when the dataset is uneven in what it observes. In this study, image observations are dense and repeated over time, but structural measurements are sparse and appear only at late stages. An earlier stage of the project showed that peak surface corrosion could be predicted from images, but that stage stopped at current-state regression. It did not use the full specimen history, it did not estimate hidden structural quantities, and it did not provide a path from current corrosion state to future threshold crossing.

The present study was designed to address that gap. Its contribution is not a claim of direct image-to-life prediction. The dataset does not contain true failure times, so that claim would not be scientifically defensible. Instead, this paper presents a staged feasibility pipeline:

raw observations → clean specimen histories →
image-derived corrosion descriptors → visible corrosion state
→ hidden structural estimate → degradation trend → proxy
remaining useful life

This is the central identity of the paper. It is a feasibility study of a multi-stage corrosion-to-condition workflow under sparse structural supervision.

The study addresses four questions:

- 1) Can visible corrosion be estimated reliably from images using interpretable features?
- 2) Can visible corrosion and specimen metadata support estimation of hidden structural quantities?
- 3) Can repeated specimen observations be summarized through useful degradation curves?
- 4) Can those curves support threshold-based proxy remaining useful life screening?

II. DATASET AND SCIENTIFIC SCOPE

The dataset contains 792 image observations from 48 specimens across 2 campaigns, covering 25 observed week values from week 0 to week 36. Each observation contains specimen identity, observation time, one corrosion image, specimen and exposure metadata, and visible corrosion labels. The

TABLE I
DATASET SUMMARY

Quantity	Value
Image observations	792
Specimens	48
Campaigns	2
Week range	0 to 36
Unique observed week values	25
Surface-labeled rows	792
Structural-labeled rows	48
Structural-label coverage	6.06%

structural targets are much rarer: 48 rows contain structural measurements, corresponding to one late-stage structural row per specimen at weeks 28 and 36 only.

A reader can easily get confused in a multi-stage pipeline unless the evidence type of each variable is made explicit. Visible corrosion is directly observed on all rows. Wire-area loss and terminal ultimate load are measured only on 48 late-stage rows. Estimated structural states, degradation trajectories, and remaining-life outputs are therefore downstream inferred quantities rather than direct observations.

The dataset supports repeated observation of visible corrosion, specimen-level time-series reconstruction, image-based estimation of visible corrosion severity, preliminary hidden-damage estimation, and threshold-based remaining-life screening. It does not support direct supervised prediction of true remaining useful life because failure times are not observed.

III. PIPELINE OVERVIEW

The workflow was designed as a chain of connected stages. Each stage has a clear input, performs one specific transformation, and produces an output that becomes the input of the next stage.

IV. STAGE-BY-STAGE METHODOLOGY

A. Dataset Reconstruction and Validation

The pipeline begins with the raw spreadsheet and the specimen images. The spreadsheet is cleaned by treating its first row as the true header row, then its columns are renamed into a consistent schema. The record identifier is parsed using a fixed pattern of the form `specimenID-YYYYMMDD-weekW`. From this identifier, the pipeline extracts `specimen_id`, `capture_date`, and `week`. The image directory is scanned independently, and every PNG file name is converted into a record identifier. The spreadsheet table and image table are then merged using those identifiers.

Additional metadata are reconstructed at this stage. Treatment codes are expanded into human-readable treatment names. Campaign labels are inferred from capture year, mesh count, salt concentration, and ageing duration. A flag is created to indicate whether structural labels are available. Sequence order is assigned within each specimen history. Duplicate record IDs and duplicate specimen-week combinations are flagged explicitly.

One engineering correction is also applied here. The raw wire-loss field behaves numerically like a fraction rather than a percentage. Therefore, when its maximum value is small enough to indicate ratio storage, it is multiplied by 100 and stored as wire-area loss percentage.

This stage creates the backbone of the entire study. Without a clean specimen-wise time-series table, later stages such as degradation fitting and remaining-life screening would not be meaningful.

B. Image Preprocessing and Corrosion Segmentation

Each aligned image is opened and converted into RGB format. If an image contains transparency, it is composited onto a white background so that all images share a consistent representation.

The pipeline then extracts the central specimen region after trimming image borders and background margins. In the current implementation, the crop removes 3% from the left, 3% from the right, 8% from the top, and 8% from the bottom. This central analysis region is used because the border areas are less informative and more likely to include non-specimen background.

After cropping, illumination is normalized in two steps. First, each RGB channel is rescaled using its 2nd and 98th percentiles. Second, the brightness channel in HSV space is enhanced using adaptive histogram equalization. A rust-likelihood map is then computed by combining red dominance over green, warm-color dominance over blue, chromatic information from the Lab space, saturation in HSV space, and closeness of hue to typical red-brown corrosion tones.

The rust-likelihood map is converted into a rust mask. The threshold is chosen from the larger of Otsu's threshold and a score-based threshold equal to mean + 0.35 times standard deviation, then clipped to a stable interval. After thresholding, the mask is cleaned by removing very small objects, filling small holes, and applying small binary opening and closing operations.

This stage converts a raw image into a corrosion-focused representation that can be measured consistently in the next stage.

C. Interpretable Feature Extraction

The pipeline converts each processed image into a structured set of quantitative corrosion descriptors. The goal is not only to measure how much rust is visible, but also where it appears, how concentrated it is, how fragmented it is, how its color shifts, and how its texture changes.

In total, the implementation extracts 91 image features. These include corrosion amount features such as rust area percentage and peak segment rust percentage; spatial features such as segment-wise corrosion and peak location; morphology features such as rust-component count and largest region area; color features in RGB, HSV, and Lab spaces; histogram summaries; and texture descriptors derived from gray-level co-occurrence matrices and local binary patterns. The specimen is divided into 12 longitudinal segments. Since the nominal

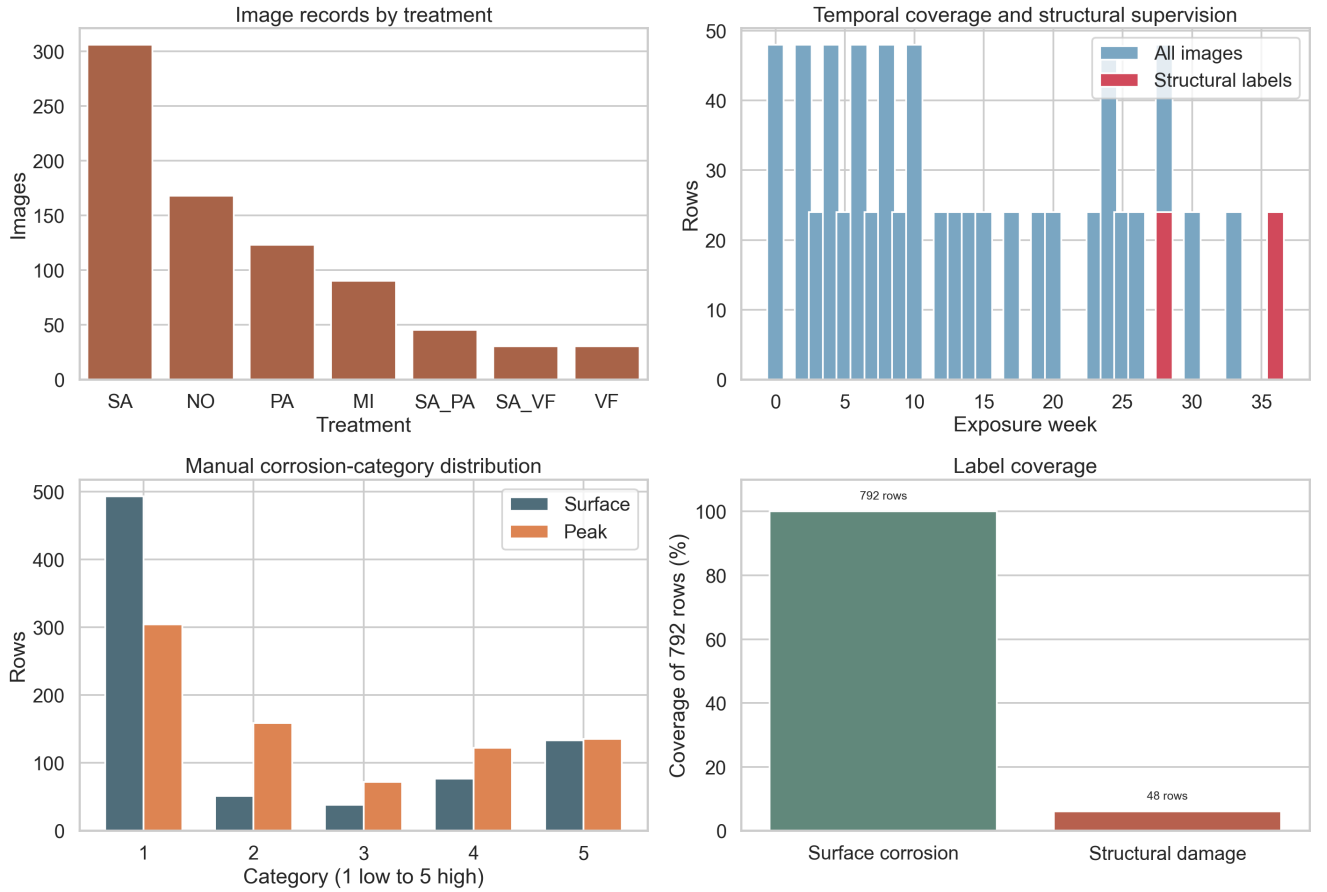


Fig. 1. Dataset overview used in the study. The upper panels summarize image-record counts by treatment and by exposure week, while the lower panels show manual corrosion-category distributions and label coverage. The lower-right panel highlights the central data imbalance that motivates the staged pipeline: surface labels exist for all 792 rows, whereas structural supervision exists for only 48 late-stage rows.

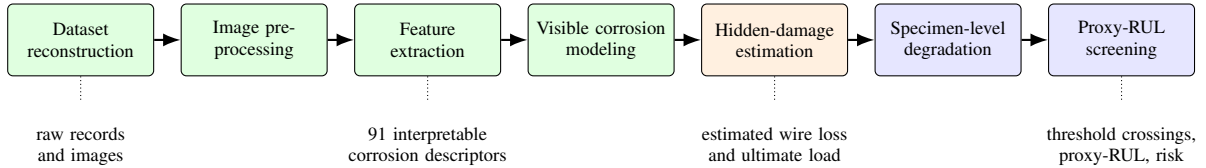


Fig. 2. Step-by-step workflow used in the study. Green stages operate on directly observed data, orange stages estimate hidden structural state, and blue stages forecast future condition and convert threshold crossing into screening outputs. The figure makes explicit that proxy-RUL is not predicted directly from images; it is derived only after corrosion modeling, hidden-damage estimation, and specimen-level degradation fitting.

specimen length is set to 24 cm, each segment corresponds to a 2 cm section, which allows the pipeline to describe where corrosion is concentrated along the specimen.

If feature extraction fails on an image, the row is not discarded. Instead, the failure is flagged and missing numeric values are later filled using median imputation. This keeps the observation table complete for downstream modeling.

D. Visible Corrosion Modeling

The first predictive stage learns visible corrosion severity directly from the image-derived feature set. Two regression targets are modeled: total visible rust percentage and peak vis-

ible rust percentage. Two categorical targets are also modeled: total-rust category and peak-rust category.

For modeling, numeric features are median-imputed and standardized. Categorical features, when present, are imputed with their most frequent value and one-hot encoded. The regression models compared are linear regression, random forest, histogram-based gradient boosting, and multilayer perceptron regression. The classification models compared are logistic regression, random forest, histogram-based gradient boosting, and multilayer perceptron classification.

The evaluation uses three split strategies: one grouped specimen holdout using `GroupShuffleSplit` with 20% test size and fixed random seed, leave-one-treatment-out, and

TABLE II
MAIN FEATURE FAMILIES

Feature family	Examples	What it describes
Corrosion amount	rust area, peak rust, rust score	Overall visible severity
Spatial distribution	segment rust, peak location	Where corrosion concentrates
Morphology	component count, region area, eccentricity	Shape and spread of corrosion
Color statistics	RGB, HSV, Lab means and variability	Discoloration and saturation shifts
Texture	contrast, homogeneity, energy, local binary patterns	Surface roughness and corrosion texture

leave-one-campaign-out. The best model for each target is selected from the grouped specimen-holdout benchmark. After model selection, the chosen estimator is refit on all available rows for that target to produce final state estimates.

This stage is the most strongly supervised part of the dataset, since visible corrosion labels exist for all rows. It validates the image features and produces a reliable corrosion-state layer.

E. Hidden Structural Damage Estimation

This stage starts from the cleaned specimen-by-time dataset, the extracted image features, and the corrosion-related columns already available in the specimen table. The implementation first merges the reconstructed specimen dataset with the image-feature table, then retains the measured visible-corrosion labels that are already present in the merged table. From that combined design table, it selects only the rows that contain structural supervision. In this dataset, that means the 48 late-stage rows where structural measurements are available.

The structural stage does not use all available columns blindly. Because only 48 structural rows exist, the implementation uses a curated subset of predictors to reduce overfitting and keep the model interpretable. The selected inputs include specimen and exposure metadata such as week, mesh count, ageing days, salt concentration, cover thickness, treatment code, campaign ID, and specimen series label; measured visible-corrosion variables such as total visible rust, peak visible rust, their severity categories, and peak-rust location; and a smaller subset of the image-derived corrosion descriptors such as rust area percentage, image-derived peak rust, rust-distribution statistics, component counts, rust-score statistics, edge density, selected texture summaries, and selected Lab and HSV color features.

This is a key implementation detail: the structural models are not trained only from predicted corrosion outputs. Since measured visible-corrosion labels already exist in the dataset, they are used directly in the structural training table.

Two structural regression targets are modeled: wire-area loss percentage and terminal ultimate load. The same tabular preprocessing framework used earlier is reused here. The regressors compared are Random Forest, Extra Trees, Histogram Gradient Boosting, and MLP Regressor. When available in the

software environment, boosted tree models such as XGBoost and CatBoost can also be included.

For each target separately, the implementation evaluates all candidate models, records the prediction metrics under the three split strategies, selects the best model based on the grouped specimen-holdout benchmark, refits that selected model on all structural rows, and then applies the refitted model to the full observation table. This produces two estimated state columns for every observation: `estimated_wire_area_loss_pct` and `estimated_ultimate_load_kn`. Estimated wire loss is clipped to 0–100%, and estimated load is clipped to non-negative values.

This stage provides the hidden structural layer needed for downstream degradation and remaining-life screening. At the same time, it is the most fragile stage of the pipeline because it is trained on only 48 rows, those rows occur only at late stages, and it must infer structural condition from weakly supervised patterns. It should therefore be interpreted as a latent-state estimation layer, not as direct internal-damage measurement.

F. Specimen-Level Degradation Modeling

This stage begins after the structural estimates have been extended to the whole dataset. The implementation merges the cleaned specimen-by-time dataset with the estimated structural states produced in the previous stage. That merged table contains, for each row, specimen identity, week, measured visible corrosion, and estimated hidden structural condition.

The degradation stage does not fit one global model for the whole dataset. Instead, it works specimen by specimen. For each specimen, it fits a separate curve for four targets: total visible rust percentage, peak visible rust percentage, estimated wire-area loss percentage, and estimated ultimate load. Since each of the 48 specimens contributes four target trajectories, the stage produces 192 fitted curves.

For each specimen, the rows are sorted by week, the observed week values and target values are extracted, a set of candidate curve families is tested, the best-fitting family is selected based on historical fit error, and the fitted curve is used to generate future values.

The implementation tries four curve families:

- linear,
- exponential,
- logarithmic,
- piecewise linear.

The piecewise-linear fit is not assumed at a fixed breakpoint. Instead, the code searches across candidate breakpoints taken from the observed weeks, excluding the first two and last two positions so that both sides of the breakpoint still contain enough data. For each candidate breakpoint, one line is fit to the left side and one line to the right side, and the piecewise-linear model with the lowest root mean squared error is selected.

For each specimen-target pair, all successful candidate curve fits are scored using RMSE, and the model with the lowest historical RMSE is selected. The pipeline stores the selected

curve family, fitted parameters, RMSE, R^2 , number of observed points, current week, current observed value, and the progression rate near the end of the fitted range.

After the best curve has been selected for a specimen-target pair, it is used to generate a forecast. The future timeline is built from the earliest observed week up to the latest observed week plus an additional forecast horizon of 104 weeks, using a step size of 0.5 weeks. Predicted percentage targets are clipped to 0–100, and predicted load is clipped to non-negative values.

For each specimen, target, and week, the forecast table stores the forecasted value, the observed value when available, whether the week belongs to the historical or forecast phase, and the selected curve family. This means the output is not just one future number, but a full time trajectory that can later be interrogated by the proxy-RUL stage.

This stage creates the time dimension needed to distinguish between specimens that are already severe but stable and specimens that are still moderate but deteriorating quickly. However, uncertainty also accumulates here. Total visible rust and peak visible rust are fitted from repeatedly observed measured values, whereas estimated wire-area loss and estimated ultimate load are fitted from model-estimated values produced by the structural stage. The structural degradation curves therefore contain both curve-fitting uncertainty and the estimation uncertainty already introduced upstream.

G. Proxy Remaining Useful Life Estimation

For each specimen, the pipeline identifies the latest observed row and uses the degradation forecasts to build a specimen-specific future table containing total visible rust, peak visible rust, estimated wire-area loss, and estimated ultimate load. A campaign-specific reference load is defined as the maximum observed ultimate load within that campaign. This reference is used to convert future load into a relative capacity margin.

A health index is also computed at each forecasted week. This index combines visible corrosion severity, estimated wire loss, and relative load reduction. In the current implementation, the health index gives 30% weight to corrosion, 40% weight to wire loss, and 30% weight to load loss.

The pipeline then searches for the earliest future threshold crossing for:

- wire loss above 20%, 25%, and 30%,
- load below 80% of the campaign reference load,
- health index below 0.35.

The reported proxy remaining useful life is the earliest finite crossing among the 25% wire-loss threshold, the load threshold, and the health threshold. If none of these thresholds is crossed within the forecast horizon, the RUL value is left missing.

A risk score is also computed from estimated RUL, wire-loss margin, load margin, and health index. That score is then converted into four risk classes: Low, Moderate, High, and Critical.

A proxy-RUL of 0 weeks means the model considers the specimen already beyond a chosen engineering threshold at the

TABLE III
EVALUATION STRENGTH ACROSS PIPELINE STAGES

Stage	Supervision level	Evaluation confidence
Visible corrosion	Dense labels on all rows	High
Hidden damage	Sparse late-stage labels only	Moderate to low
Degradation fitting	Observed surface + estimated structural states	Moderate
Proxy-RUL	No direct failure-time supervision	Screening only

latest observation. It does not mean observed immediate physical failure. This final stage therefore provides an engineering screening output rather than a true lifetime prediction.

V. EVALUATION PROTOCOL

Because each specimen appears repeatedly across time, train-test splitting must keep specimen histories separated. Otherwise, the model could train on one week of a specimen and be tested on another week of that same specimen.

The benchmark bundle contains three evaluation views:

- 1) grouped specimen holdout, which is the main benchmark and uses one grouped specimen split with fixed random seed;
- 2) leave-one-treatment-out, which checks transfer across treatment groups;
- 3) leave-one-campaign-out, which checks transfer across the two campaigns.

The main headline numbers come from the grouped specimen-holdout benchmark. The harder split results are used to judge robustness.

VI. RESULTS

A. Visible Corrosion Modeling

The visible-corrosion stage was the strongest part of the pipeline. In the grouped specimen-holdout benchmark, the best model for total visible rust was a Random Forest with MAE = 1.38, RMSE = 3.14, and $R^2 = 0.646$. The best model for peak visible rust was an MLP regressor with MAE = 4.21, RMSE = 6.23, and $R^2 = 0.784$.

This also improved the earlier peak-rust baseline, which had MAE = 5.63, RMSE = 9.26, and $R^2 = 0.524$.

The category models were weaker, but still useful: total-rust category reached accuracy = 0.830 and macro-F1 = 0.573, while peak-rust category reached accuracy = 0.591 and macro-F1 = 0.531.

B. Hidden-Damage Modeling

The hidden-damage stage showed some learnable signal, but much less strongly than the visible-corrosion stage. In the grouped specimen-holdout benchmark, the best model for

TABLE IV
MAIN GROUPED-HOLDOUT RESULTS

Task	Best model	MAE	RMSE	R^2
Total visible rust (%)	Random Forest	1.38	3.14	0.646
Peak visible rust (%)	MLP Regressor	4.21	6.23	0.784
Wire-area loss (%)	Random Forest	3.90	5.00	0.310
Ultimate load (kN)	Random Forest	0.121	0.168	0.385

TABLE V
SELECTED DEGRADATION CURVE FAMILIES

Target	Exponential	Piecewise linear
Total visible rust	8	40
Peak visible rust	4	44
Estimated wire-area loss	0	48
Estimated ultimate load	0	48

wire-area loss was Random Forest with MAE = 3.90, RMSE = 5.00, and $R^2 = 0.310$. The best model for ultimate load was Random Forest with MAE = 0.121 kN, RMSE = 0.168 kN, and $R^2 = 0.385$.

These structural results are best interpreted as feasibility evidence rather than mature predictive performance.

C. Harder Holdouts

The stricter holdouts made the evidence hierarchy clearer. For visible corrosion, grouped specimen holdout remained the strongest setting, while treatment and campaign holdouts reduced performance but still retained some useful signal. For hidden damage, performance degraded sharply, and mean R^2 values became strongly negative in the harder settings. This means the structural stage did not transfer robustly across more difficult splits.

D. Degradation Modeling

The degradation stage produced 192 fitted curves and 52,416 forecast rows. Most selected curve families were piecewise linear.

This result is useful descriptively: it shows that specimen-level trend fitting is possible, but it also shows that the resulting curves are pragmatic summaries rather than physical degradation laws. Fig. 3 illustrates both the average peak-rust progression across the dataset and representative specimen-wise trajectories used for forward screening.

E. Proxy Remaining Useful Life

The final remaining-life layer produced screening outputs rather than true lifetime predictions. Across the 48 specimens, 33 had finite proxy-RUL estimates, 15 did not cross a threshold within the forecast horizon, 23 had proxy-RUL equal to 0 weeks, mean proxy-RUL was 5.42 weeks, median proxy-RUL was 0.00 weeks, and maximum proxy-RUL was 55.5 weeks. The resulting risk classes were 20 Low, 8 Moderate, 20 High, and 0 Critical.

TABLE VI
PROXY-RUL SUMMARY

Quantity	Value
Specimens	48
Finite proxy-RUL values	33
No crossing within horizon	15
Zero-week proxy-RUL	23
Mean proxy-RUL	5.42 weeks
Median proxy-RUL	0.00 weeks
Max proxy-RUL	55.5 weeks

Fig. 4 shows the final distribution of screening classes. In this run, the pipeline separated the cohort into low-, moderate-, and high-risk groups, while no specimen reached the critical band.

VII. DISCUSSION

The strongest validated result of the study is visible-corrosion modeling. That stage is well supported because the labels are dense, the image evidence is direct, and the grouped specimen benchmark shows substantial predictive power.

The hidden-damage stage is more limited. It shows that some structural signal is present, but the strong degradation under treatment-holdout and campaign-holdout evaluation shows that this part of the pipeline is still constrained by sparse and late-stage supervision.

The degradation stage is useful, but should not be over-interpreted. Good historical fit does not automatically imply reliable future structural forecasting, especially when the structural trajectories themselves depend on model-estimated hidden states.

The proxy-RUL stage is therefore best understood as a downstream engineering screening layer. It is useful, but only within the limited scope allowed by the data.

The main achievement of the study is not a single score. It is the construction of a coherent corrosion-to-condition workflow that respects the structure of the data: direct evidence first, sparse inference second, empirical forecasting third, screening output last.

VIII. THREATS TO VALIDITY

Several limitations directly affect interpretation:

- 1) sparse structural supervision, because the structural targets are observed only once per specimen;
- 2) no true failure times, which makes supervised remaining-life prediction impossible;
- 3) uneven evidence across stages, because the visible-corrosion stage is well validated while the later stages rely on weaker supervision and propagated model uncertainty;
- 4) empirical degradation curves, which are useful trend summaries but not mechanistic laws;
- 5) weak transfer in the structural stages, because hidden-damage models degrade strongly under harder holdouts.

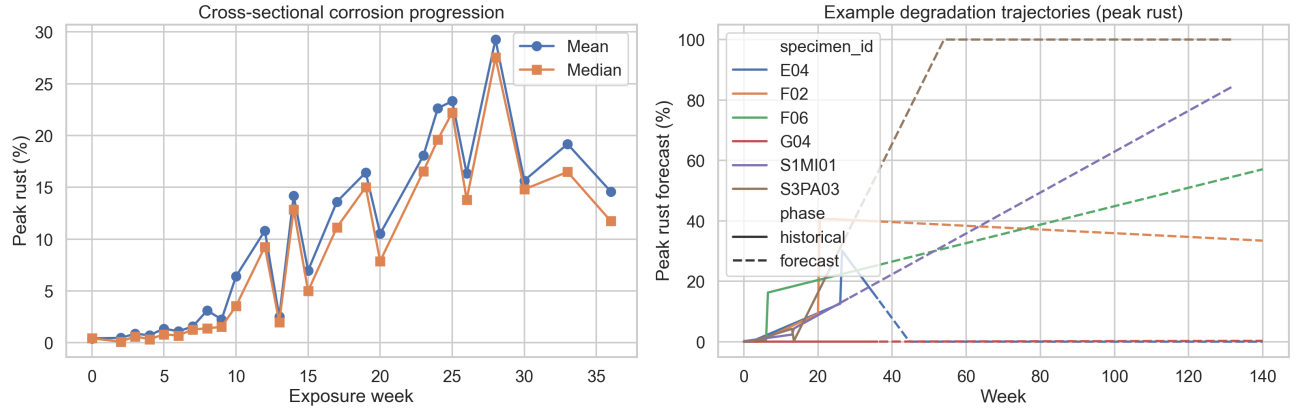


Fig. 3. Representative degradation behavior used by the pipeline. The left panel shows the average and median peak-rust progression across exposure weeks, while the right panel shows example specimen-wise historical and forecast trajectories used for threshold-based screening. Together, the panels illustrate why degradation fitting is treated here as an empirical trend model rather than as a single global deterioration law.

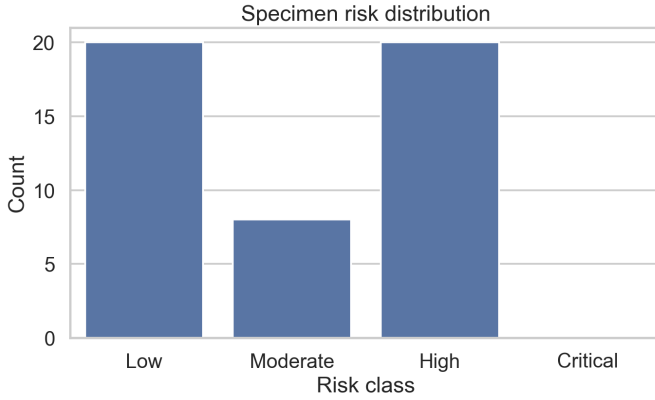


Fig. 4. Distribution of the final risk classes assigned by the proxy-RUL screening stage. In this run, the cohort separates into low-, moderate-, and high-risk groups, while no specimen enters the critical band under the chosen screening thresholds.

IX. CONCLUSION

This paper presents an interpretable multi-stage feasibility pipeline rather than a final remaining-life predictor. The study reconstructs specimen histories, extracts 91 interpretable corrosion features, estimates visible corrosion and hidden structural quantities, fits specimen-level degradation curves, and converts threshold crossing into proxy remaining useful life.

The strongest evidence lies in visible-corrosion modeling, where the pipeline improved the shared peak-rust task and performed well under grouped specimen holdout. Hidden-damage estimation remained feasible but much weaker, making sparse structural supervision the main bottleneck. Degradation fitting and proxy-RUL were useful as downstream screening components, but not as validated lifetime prediction tools.

The most defensible conclusion is therefore:

Repeated corrosion images can support an interpretable corrosion-to-condition pipeline ending in proxy remaining useful life, but the later structural

stages remain limited by sparse and late-stage supervision.